

1. Administrative & Study Identification

Full Study Title: A Study of Zilovetamab Vedotin (MK-2140) in Combination With Standard of Care in Participants With Relapsed or Refractory Diffuse Large B-Cell Lymphoma (rrDLBCL) (MK-2140-003) **Short Title/Acronym:** waveLINE-003 / MK-2140-003 **Protocol Number:** MK-2140-003 **NCT Number:** NCT05139017 **Sponsor Name:** Merck **Investigational Product:** Zilovetamab Vedotin (ZV / MK-2140) **Phase of Development:** Phase 2, Phase 3 **Medical Monitor:** Merck Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety and efficacy of zilovetamab vedotin (ZV) in combination with standard of care options for the treatment of rrDLBCL, and to determine if ZV in combination with rituximab, gemcitabine, and oxaliplatin (R-GemOx) is superior to R-GemOx alone with respect to progression-free survival (PFS). **Secondary Objectives:** To evaluate the Objective Response Rate (ORR), Duration of Response (DOR), Overall Survival (OS), Safety, and Pharmacokinetics.

2.2 Background & Rationale Consensus treatment guidelines are unavailable for patients with diffuse large B-cell lymphoma (DLBCL) whose disease progresses after first-line therapy. Outcomes for relapsed/refractory (R/R) DLBCL remain poor. The receptor tyrosine kinase-like orphan receptor 1 (ROR1) is a transmembrane protein overexpressed in multiple hematologic malignancies. Zilovetamab vedotin (ZV) is an antibody-drug conjugate targeting ROR1 that has shown promising efficacy and tolerability in hematologic malignancies, prompting this dose confirmation and efficacy expansion study.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active-control (Standard of care R-GemOx) **Blinding:** Open-label (with Blinded Independent Central Review [BICR]) **Randomization:** Randomized (1:1 ratio)

3.2 Planned Enrollment & Duration Target \$N\$: 290 participants (Enrollment adjusted due to Cohort B discontinuation) **Number of Sites:** Multisite (Global locations including the UK and US) **Estimated Study Start:** December 2021 **Estimated Primary Completion:** Ongoing/Recruiting

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years. 2. Histologically confirmed Diffuse Large B-Cell Lymphoma (DLBCL) per WHO classification. 3. Relapsed or refractory disease (rrDLBCL) after ≥ 1 line of prior therapy. 4. Ineligible for or have failed autologous stem-cell transplant (ASCT) and have post-chimeric antigen receptor T (post-CAR-T) cell therapy failure or are ineligible for CAR-T cell therapy. 5. Radiographically measurable disease per Lugano Response Criteria and an Eastern Cooperative Oncology Group (ECOG) performance status of 0 to 2 within 7 days prior to study treatment initiation. 6. Has adequate organ function. 7. Is able to provide new or archival tumor tissue sample not previously irradiated.

4.2 Exclusion Criteria 1. Diagnosis of primary mediastinal B-cell lymphoma (PMBCL). 2. Known active central nervous system (CNS) lymphoma involvement (prior CNS involvement eligible only if in radiographic, cytological, and clinical remission). 3. Ongoing corticosteroid therapy. 4. Received a live or live-attenuated vaccine within 30 days before the first dose of study intervention. 5. Participating in or has participated in a study of an investigational agent or has used an investigational device within 4 weeks before the first dose. 6. Has received a solid organ transplant at any time. 7. Has clinically significant (ie, active) cardiovascular disease or serious cardiac arrhythmia requiring medication. 8. Has ongoing graft-versus-host disease (GVHD) of any grade, or is receiving treatment for their GVHD. 9. Has clinically significant pericardial or pleural effusion. 10. Has ongoing Grade >1 peripheral neuropathy or a demyelinating form of Charcot-Marie-Tooth disease. 11. Has a history of a second malignancy, unless potentially curative treatment has been completed with no evidence of malignancy for 2 years. 12. Has an active infection requiring systemic therapy, a known history of HIV infection, or a known active Hepatitis C virus infection. 13. Has a known psychiatric or substance abuse disorder that would interfere with the participant's ability to cooperate with the requirements of the study. 14. Has received prior radiotherapy within 4 weeks of start of study intervention without full recovery from toxicities.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Zilovertamab vedotin (ZV) at the recommended phase 2 dose (RP2D) of 1.75 mg/kg combined with R-GemOx (Rituximab, Gemcitabine, Oxaliplatin). *(Note: Cohort B evaluating ZV + Bendamustine Rituximab [BR] was discontinued per protocol amendment 4).* **Route of Administration:** Intravenous infusions. **Duration of Treatment:** Up to 6 cycles of combination therapy.

5.2 Concomitant & Prohibited Therapy Permitted: Granulocyte-colony stimulating factor (G-CSF) to help normal white blood cells recover from chemotherapy. **Prohibited:** Live vaccines, systemic corticosteroids (unless specified), and other investigational agents.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Confirmation of relapsed/refractory status, Baseline scans
Baseline	Day 0	Randomization, First Dose of ZV + R-GemOx (or R-GemOx alone)
Interim	Every 12 weeks	Disease response assessments by CT and PET scans, Safety Labs
End of Study	Follow-up	Final Vital Signs, Exit Interview, Disease progression or study discontinuation monitoring

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints Definition: - Progression-Free Survival (PFS) per Lugano 2014 criteria (Hypothesis testing). - Number of participants who experienced dose-limiting toxicities (DLTs) in Part 1. - Number of participants who experienced an adverse event (AE). - Number of participants who discontinued study treatment due to an AE.

Measurement Method: Evaluated by Blinded Independent Central Review (BICR) utilizing CT and PET scans every 12 weeks for PFS. CTCAE, Version 5.0 will be used to grade the severity of AEs and DLTs.

7.2 Secondary & Exploratory Endpoints - Objective Response Rate (ORR) - Duration of Response (DOR) - Overall Survival (OS) - Pharmacokinetics

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring for treatment-related AEs (TRAEs). Key events monitored include neutropenia, anemia, nausea, and diarrhea (which were commonly reported Grade ≥ 3 TRAEs). **Serious Adverse Events (SAEs):** Standard 24-hour reporting. **Data Monitoring Committee (DMC):** Utilized for the dose confirmation phase (mTPI design) and safety run-in.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for efficacy. Safety Set for toxicity evaluations. Per protocol amendment 4, no efficacy outcome analysis and hypothesis testing will be conducted for Cohort B. **Sample Size Justification:** 290 participants initially designed to power the superiority hypothesis of PFS for the investigational regimens vs. active comparators. **Handling of Missing Data:** Standard oncology guidelines for censoring PFS data at the last evaluable tumor assessment.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Onsite and remote monitoring visits for source data verification. **Audit Trail:** eCRF locking, query resolution via data clarification forms, and Independent Central Review (BICR) for objective radiological assessments.

11. Study Identifiers & Governance

Primary ID: MK-2140-003 **Registry Number:** NCT05139017 **Posting ID:** 153223240357976234 **Sponsor/Lead Organization:** Merck **Study Title:** waveLINE-003 **Official Regulatory Title:** A Phase 2/3 Multicenter, Open-label, Randomized, Active-Control Study of Zilovetamab Vedotin (MK-2140) in Combination With Standard of Care in Participants With Relapsed or Refractory Diffuse Large B-Cell Lymphoma (waveLINE-003)

12. Clinical Framework (Disease Specific)

Primary Condition: Relapsed or Refractory Diffuse Large B-Cell Lymphoma (rrDLBCL) **Diagnosis Threshold:** Histologically confirmed DLBCL, ineligible for/failed ASCT and CAR-T **Medical Methodology:** ROR1-targeted ADC + Standard of Care Chemotherapy (R-GemOx) **Optimization Target:** Progression-Free Survival (PFS) and Objective Response Rate (ORR)

13. Study Procedures & Methodology

Management Pathway: Standard oncology clinical pathway for R/R DLBCL **Education Protocol (HCP):** Oncology protocol training / Site initiation visits **Education Protocol (Patient):** Informed consent and treatment counseling **Quality Audit Mechanism:** Blinded Independent Central Review (BICR) for disease response

14. Observation & Follow-Up Schedule

Total Duration: Until disease progression or unacceptable toxicity **Onsite Visit Cadence:** Every 3 weeks per treatment cycle / Imaging every 12 weeks **Remote Monitoring Frequency:** Q4W or per protocol **Checklist Review Interval:** Every 12 weeks (Q12W) corresponding with scans

15. Sign-off & Approval

Role	Name	Signature	Date		----	----	----	----
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]					